PH4201: Adv. Optics Lab Report

Measurement of Verdet Constant and Verification of Malus' Law

Debayan Sarkar
Roll No: 22MS002

Sabarno Saha
Roll No: 22MS037

Diptanuj Sarkar
Roll No: 22MS038

Date: 17 April 2026

1. Aim

- To measure the Verdet Constant by observing how the plane of polarisation varies as we increase the magnetic field in a solenoid.
- To verify Malus's law

2. Materials required

1. Laser source
2. Electromagnet
3. Photodiode Detector
4. Linear Polarisers

3. Theory

3.1. Linearly Polarised Light as a superposition of Circularly Polarised Components

Linearly polarised light travelling along the x-direction polarised along the y-direction can be written as a superposition of left and right circularly polarised light as:

$$\boldsymbol{E}(x,t) = \boldsymbol{E}_r(x,t) + \boldsymbol{E}_l(x,t) \tag{3.1}$$

where,

$$\begin{aligned} \boldsymbol{E}_r(x,t) &= \begin{pmatrix} 0 \\ \frac{A}{2}\cos\left(\omega\left(t - \frac{nx}{c}\right)\right) \\ \frac{A}{2}\sin\left(\omega\left(t - \frac{nx}{c}\right)\right) \end{pmatrix} \\ \boldsymbol{E}_l(x,t) &= \begin{pmatrix} 0 \\ \frac{A}{2}\cos\left(\omega\left(t - \frac{nx}{c}\right)\right) \\ -\frac{A}{2}\sin\left(\omega\left(t - \frac{nx}{c}\right)\right) \end{pmatrix} \end{aligned} \tag{3.2}$$

Here n is the refractive index for each of these polarisations. However, it's possible for the polarisations to be different for each of the polarisation states. Let's say the refractive indices for the RCP and LCP states are n_r and n_l. Then, at some $x = x_0$, the electric field is given by,

$$\boldsymbol{E}(x,t) = \begin{pmatrix} 0 \\ A\cos\big(\omega\big(t-(n_r+n_l)\frac{x_0}{2c}\big)\big)\cos\big(\omega(n_r-n_l)\frac{x_0}{2c}\big) \\ A\cos\big(\omega\big(t-(n_r+n_l)\frac{x_0}{2c}\big)\big)\sin\big(\omega(n_r-n_l)\frac{x_0}{2c}\big) \end{pmatrix} \tag{3.3}$$

So the plane of polarisation now makes an angle ψ with the y-axis, which is given by

$$\psi = \arctan\left(\frac{E_z}{E_y}\right) = \frac{\omega(n_r-n_l)x_0}{2c} = \frac{\pi(n_r-n_l)D}{\lambda} \tag{3.4}$$

where we have replaced x_0 with D to denote the distance travelled by light in this bifringent material. λ is the wavelength of the beam. As we can see, if $D \gg \lambda$, then even for small values of $n_r - n_l$ we can see significant rotations in the plane of polarisation of the input light beam.

3.2. Magnetic Circular Birefringence

For a Solenoid with Magnetic Field B and length L, the rotation of the plane of polarisation is given by

$$\psi = VLB \tag{3.5}$$

where V is the Verdet Constant. Using this equation, by measuring ψ for different values of B, we can measure the value of the Verdet Constant for a given solenoid.

3.3. Malus' Law

Consider a Linearly polarised light beam passing through a linear polariser that makes an angle θ with the plane of polarisation of the incident wave. Then, the Electric Field of the transmitted beam is given by:

$$E_{\text{out}} = E_{\text{in}}\cos(\theta) \tag{3.6}$$

Since Intensity $I \propto E^2$, we then have

$$I_{\text{out}} = I_{\text{in}}\cos^2(\theta) \tag{3.7}$$

This equation is Malus' Law. We can verify this by measuring the intensity of the trasmitted beam through the second polariser for different values of θ.

4. Data

We obtained 3 sets of data for the determination of the Verdet Constant, and two sets of data for verification of Malus' Law. The data tables have been included in the supplmentary section at the end of the report.

Here are a few important parameters for our experiment:

- $\lambda = 650$ nm
- $L = 10cm$
- Number of turns per unit length = 167.2 turns/cm

5. Results

5.1. Measurement of Verdet Constant

The values of Verdet constant obtained from each dataset were calculated by fitting the measured data linearly. The obtained values have been shown in the plot below

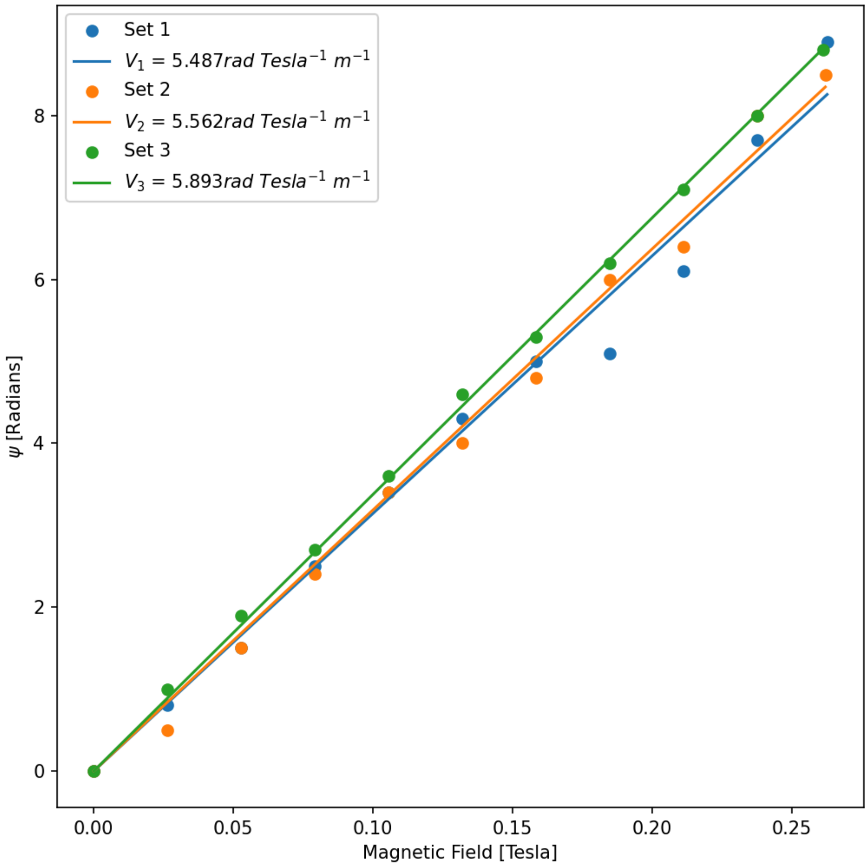

Figure 1: Plot ψ vs B to obtain the Verdet Constant V for 3 different sets of data

The average value of Verdet Constant obtained from the three sets is

$$V_{\text{avg}} = 5.647 \text{ rad Tesla}^{-1} m^{-1} \tag{5.1}$$

5.2. Verification of Malus' Law

We did a linear fit on the loglog plot of Magnetic field vs Intensity as shown in the figure below

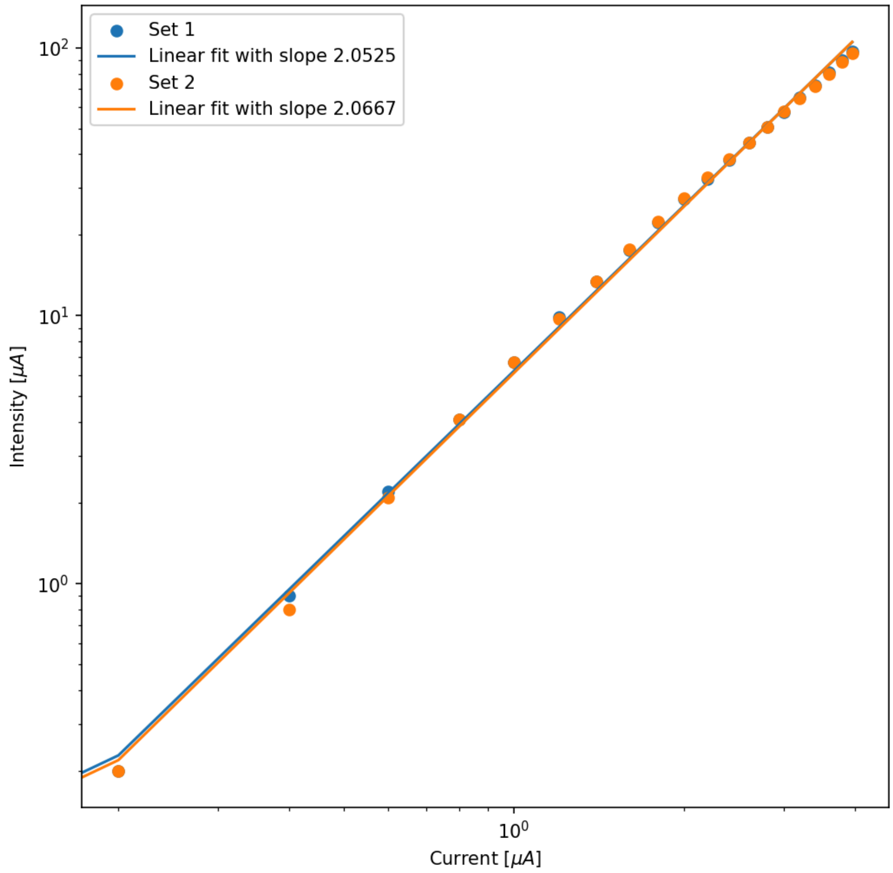

Figure 2: loglog plot of the Current through the solenoid and the detected Intensity

As we can see the average slope for the two data sets is $m_{\text{avg}} = 2.0596$, which shows quadtratic behaviour, consitent with small angle behaviour of the cosine squared function.

6. Error Analysis

6.1. Error in Verdet Constant

From the standard deviations for the linear fits, the error for the average value of V was obtained to be

$$\begin{aligned} \Delta V_{\text{avg}} &= \frac{1}{3}\sqrt{(\Delta V_1)^2 + (\Delta V_2)^2 + (\Delta V_3)^2} \\ &= 0.007 \text{ rad Tesla}^{-1} m^{-1} \end{aligned} \tag{6.1}$$

From the working formula, the expression for the maximum permissible error in V is given by

$$\begin{aligned}\frac{\Delta V}{V} &= \left|\frac{\Delta B}{B}\right| + \left|\frac{\Delta \psi}{\psi}\right| \\ &= \left|\frac{\Delta I}{I}\right| + \left|\frac{\Delta \psi_{\text{deg}}}{\psi_{\text{deg}}}\right| \\ &= \frac{0.01}{0.4} + \frac{0.1}{0.5} = 0.225\end{aligned} \tag{6.2}$$

Hence, accounting for the maximum permissible error, the value we report the value of the Verdet Constant for the given setup to be

$$V = (7.096 \pm 1.270) \text{ rad Tesla}^{-1} m^{-1} \tag{6.3}$$

6.2. Error in the Malus' Law exponent

We can obtain the error in the average slope of Malus' Law can be calculated to be

$$\Delta m_{\text{avg}} = \frac{1}{2}\sqrt{(\Delta m_1)^2 + (\Delta m_2)^2} \approx 0.0003 \tag{6.4}$$

7. Sources Of Error

Sources of error for this experiment can be:

1. Non-uniform Magnetic Field: The experiment assumes that the magnetic field is uniform throughout the material. In reality, it is not perfectly uniform, which introduces error in the measurements.
2. Non-monochromatic Light: If the light is not monochromatic, different frequency components experience different rotations (due to different responses of RCP and LCP modes). Hence, the incident light should ideally be monochromatic.
3. Thermal Fluctuation: The Verdet constant and material properties can be temperature dependent. Thermal fluctuations can therefore lead to experimental errors.
4. Alignment Issue: If the laser is not properly aligned to pass horizontally through the medium, some intensity may be lost. This can lead to incorrect readings in the photodetector.
5. Imperfect Polariser: If the polarisers are not ideal, the light may not be perfectly polarised. This introduces noise and affects the accuracy of angle measurements
6. Reported Values: The length of the solenoid and the number of turns per unit length are also important parameters in the working formula. These values are provided by the manufacturer and may have some error associated with them, which can affect the final value of the Verdet constant obtained from the experiment.
7. Verification Of Malus' Law: The verification of Malus' law done by us, is technically not perfect, since we have not measured over a full period of the cosine squared function. Hence, we are only verifying the small angle behaviour of the cosine squared function, which is a quadratic dependence.

8. Discussion and Conclusion

In this experiment, we investigated the Faraday effect in flint glass and verified Malus' law through intensity measurements. It was observed that the angle of rotation of the plane of polarisation increases linearly with the applied magnetic field, as expected theoretically.

The Verdet constant was experimentally determined to be

$$V = (5.647 \pm 1.270) \text{ rad Tesla}^{-1} m^{-1} \tag{8.1}$$

In the second part, the variation of transmitted intensity with current was studied. The log-log analysis showed a power-law dependence with exponent $\approx 2.0596 \pm 0.0003$, confirming the quadratic relationship predicted by Malus' law under small-angle approximation.

Thus, the experiment successfully validates both the linear dependence of Faraday rotation on magnetic field and the quadratic dependence of transmitted intensity on the rotation angle

9. Supplementary Data

9.1. Measurement of Verdet Constant

Set 1: Measurement of Verdet Constant

Current (A)	Magnetic Field (Tesla)	θ [degrees]			$\Delta\theta$ [degrees]
		MSD	VSD	Total	
0	0	113	0	113	0
0.4	0.02640316569	112	2	112.2	0.8
0.8	0.05280633139	111	5	111.5	1.5
1.2	0.07920949708	110	5	110.5	2.5
1.6	0.1056126628	109	6	109.6	3.4
2	0.1320158285	108	7	108.7	4.3
2.4	0.1584189942	108	0	108	5
2.8	0.1848221599	107	9	107.9	5.1
3.2	0.2112253256	106	9	106.9	6.1
3.6	0.2376284912	105	3	105.3	7.7
3.98	0.2627114987	104	1	104.1	8.9

Set 2: Measurement of Verdet Constant

Current (A)	Magnetic Field (Tesla)	θ [degrees]			$\Delta\theta$ [degrees]
		MSD	VSD	Total	
0	0	113	0	113	0
0.4	0.02640316569	112	5	112.5	0.5
0.8	0.05280633139	111	5	111.5	1.5
1.2	0.07920949708	110	6	110.6	2.4
1.6	0.1056126628	109	6	109.6	3.4
2	0.1320158285	109	0	109	4
2.4	0.1584189942	108	2	108.2	4.8
2.8	0.1848221599	107	0	107	6
3.2	0.2112253256	106	6	106.6	6.4
3.6	0.2376284912	105	0	105	8
3.97	0.2620514195	104	5	104.5	8.5

Set 3: Measurement of Verdet Constant

Current (A)	Magnetic Field (Tesla)	θ [degrees]			$\Delta\theta$ [degrees]
		MSD	VSD	Total	
0	0	113	3	113.3	0
0.4	0.02640316569	112	3	112.3	1
0.8	0.05280633139	111	4	111.4	1.9
1.2	0.07920949708	110	6	110.6	2.7
1.6	0.1056126628	109	7	109.7	3.6
2	0.1320158285	108	7	108.7	4.6
2.4	0.1584189942	108	0	108	5.3
2.8	0.1848221599	107	1	107.1	6.2
3.2	0.2112253256	106	2	106.2	7.1
3.6	0.2376284912	105	3	105.3	8
3.96	0.2613913404	104	5	104.5	8.8

9.2. Verification of Malus' Law

Current (A)	Intensity(μA) Set 1	Intensity (μA) Set 2
0	0	0
0.2	0.2	0.2
0.4	0.9	0.8
0.6	2.2	2.1
0.8	4.1	4.1
1	6.7	6.7
1.2	9.9	9.7
1.4	13.4	13.4
1.6	17.6	17.7
1.8	22.2	22.4
2	27.3	27.5
2.2	32.4	32.8
2.4	38.1	38.3
2.6	44.1	44.3
2.8	50.7	50.5
3	57.6	57.8
3.2	65.2	64.7
3.4	72.6	71.8
3.6	81	79.8
3.8	90	88.4
3.96	96.6	95.1